A-A-53789
30 December 1988

COMMERCIAL ITEM DESCRIPTION

TEST LUNG

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a portable pneumo-mechanical device designed to quantitatively simulate typical adult pulmonary physiology suitable for testing and calibration of ventilation apparatii and as a training aid for respiratory therapy personnel.

Salient characteristics.

The test lung shall be capable of quantitatively simulating adult human pulmonary physiology. Lung compliance and resistance shall be adjustable.

Compliance. Compliance shall be continuously adjustable from 0.01 liter/cm H_2O to 0.15 liter/cm H_2O. Compliance accuracy shall be within 5 percent.

Resistance. There shall be a minimum of four selectable resistances. Resistance shall adjust from 2.3 to 62.9 cm H_2O/liter/second measured at 1.0 liter/second, flowrate. Accuracy shall be within 5 percent.

Lungs. The test lung unit shall have two "lungs" with a maximum tidal volume of at least 2.0 liters per lung. While at rest, each lung shall retain approximately 918 milliliters of residual gas. The unit shall have provisions for draining moisture accumulated within the lungs without disassembly of the unit. Tidal volume shall be indicated in 100 ml increments calibrated to within 3 percent and pressure corrected to 1 atmosphere. The test lung shall accommodate the use of oxygen concentration measuring devices in each "lung".

AMSC/NA FSC 6910

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Pressure. The test lung shall be equipped with a range of -10 to 120 (+/- 2 percent) cm H_2O. The monometers shall be capable of measuring the pressure within each lung independently and simultaneously. The test lung shall provide a means of measuring pressures at other points within the airway, which shall include the oropharyngeal upper bronchial. All manometers shall permit parallel tie-in of other pressure sensors/recorders from the manometers. The test lung shall provide a means of producing a sub-ambient differential pressure within the lungs and/or airway.

Carrying case. The test lung shall be housed in an ABS plastic carrying case. The case shall have rubber or metal feet to protect the case while stored and to stabilize the unit while in use.

Physical size. The weight of entire unit shall not exceed 40 pounds (18.2 Kg). The entire unit including carrying case shall be approximately 20-1/4 inches (51.4 cm) high by 25-1/4 inches (64.8 cm) wide by 8-1/2 inches (21.6 cm) deep.

Documentation. Two sets of manuals shall be provided with each unit. Each set shall include two separate manuals. One manual shall contain user/operator instructions and one manual shall contain technical information for biomedical technician use. The manuals shall include the topics specified below. Topics a and b or d, e, and f may be combined.

a. User/operator manual. Shall include an index for user/operator reference. Shall include step-by-step instructions for starting, operating, and stopping the system. Readings and results to be expected from properly adjusted and operated equipment shall be supplied calling out all operating controls, indicators, and displays. All references to controls, indicators, and displays shall follow the specific description used in the system.

b. User maintenance manual. Shall include an index for user/operator reference. Shall include performance checks which may normally be performed by the operator.

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c. Service manual that includes preventive maintenance, calibration and repair procedures. Shall include an index for biomedical technician reference. Shall be sufficient in detail to instruct a technician the proper method to remove and install (for repair) each component and sub-component which is normally site removable for repair, and to instruct a biomedical technician the proper method to repair each component/sub-component which is normally repaired at site. Shall include maintenance adjustments and procedures to be performed periodically for purpose of preventing failure or impairment of equipment operation. Information shall be included to permit a technician to locate trouble and make repairs or adjustments to the equipment.

d. Parts price list. Shall include all site replaceable parts except common fasteners. Shall include price, manufacturer's name and part no., address, and telephone number.

e. Wiring schematics. Shall include all wiring schematics of the system for use by biomedical repair technician.

Workmanship. The unit shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One lung test set with carrying case, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for tests. Sampling for testing shall be conducted in accordance with MIL-STD-105. Testing shall be conducted at inspection level II with an AQL of 2.5 defects per hundred units. The unit of product for test purposes shall be one test lung unit.

Tests. Testing shall be conducted to determine compliance with specified requirements. Testing shall include the following:

a. The compliance and resistance parameters shall be checked to insure maximum and minimum limits are attainable. Compliance is the measured volume required to build a pressure of 1 cm/H_2O. and expressed in liters/cm H_2O. Resistance is the measured pressure drop across the resistor needed to maintain volume and expressed in cm H_2O/liter/second.

b. Tidal volume. Tidal volume shall be checked to insure that the indicated volume is within 3 percent of true volume corrected to one atmosphere. This test may be performed using a known volume from a calibrated source.

c. The manometers shall be checked to insure that they are capable of indicating from -10 to +120 cm H_2O and that they are accurate within 2 percent. This test shall be performed using a manometer that is certified as accurate within 1 percent.

d. The test lung shall be tested to insure that it is capable of producing a sub-ambient pressure of -10 cm H_2O within the lungs and breathing circuit. This may be verified using the manometers of the test lung after they are checked for accuracy or by using a separate manometer.

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Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6910-01-167-6641	TEST LUNG, RESPIRATORY TRAINING, Portable

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6910-0015

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